

Vehicle paramter	Variable	Value	Units
Gravitational Acceleration	g	9.81	ms^{-2}
Vehicle Total Mass	m_t	241.7	kg
Vehicle Sprung Mass	m_s	180.9	kg
Sprung mass C.G. to Front Axle Distance	w_f	0.817	m
Sprung mass C.G. to Rear Axle Distance	w_r	0.707	m
Front Track Width	w_f	1.34	m
Rear Track Width	w_r	1.34	m
Sprung mass C.G. to Ground Height	h	0.23	m
Vehicle C.G. to Front Axle	L_f	0.64	m
Vehicle C.G. to Rear Axle	L_r	0.88	m
Pitch Moment of Inertia	I_p	64.46	kgm^2
Roll Moment of Inertia	I_r	15.34	kgm^2
Yaw Moment of Inertia	J_z	70.38	kgm^2
Front Left Unsprung Mass	m_{uFL}	7.4	kg
Front Right Unsprung Mass	m_{uFR}	7.4	kg
Rear Left Unsprung Mass	m_{uRL}	7.8	kg
Rear Right Unsprung Mass	m_{uRR}	7.8	kg
Wheel Radius	r_w	0.406	m
Wheel Moment of Inertia	I_w	3.033	m
Front Left Suspension Stiffness	k_{sFL}	19.3	N/m
Front Right Suspension Stiffness	k_{sFR}	19.3	N/m
Rear Left Suspension Stiffness	k_{sRL}	24.8	N/m
Rear Right Suspension Stiffness	k_{sRR}	24.8	N/m
Front Left Suspension Damping Coefficient	c_{sFL}	1437.611	Ns/m
Front Right Suspension Damping Coefficient	c_{sFR}	1437.611	Ns/m
Rear Left Suspension Damping Coefficient	c_{sRL}	1644.42	Ns/m
Rear Right Suspension Damping Coefficient	c_{sRR}	1644.42	Ns/m
Front Left Tire Stiffness	k_{tFL}	113832.443	N/m
Front Right Tire Stiffness	k_{tFR}	113832.443	N/m
Rear Left Tire Stiffness	k_{tRL}	113832.443	N/m
Rear Right Tire Stiffness	k_{tRR}	113832.443	N/m
Coefficient of Lift	C_L	3.18	N/A
Coefficient of Drag	C_D	1.36	N/A
Frontal Area	A_F	1.106	m^2
Air Density	ρ	1.204	kgm^{-3}
Coefficient of Rolling Friction	$F_{rolling}$	.02	N/A
Gear Ratio	gr	13	N/A
Peak Motor Torque	$tq_{motorPk}$	21	Nm

Table 1: Vehicle Model Parameters